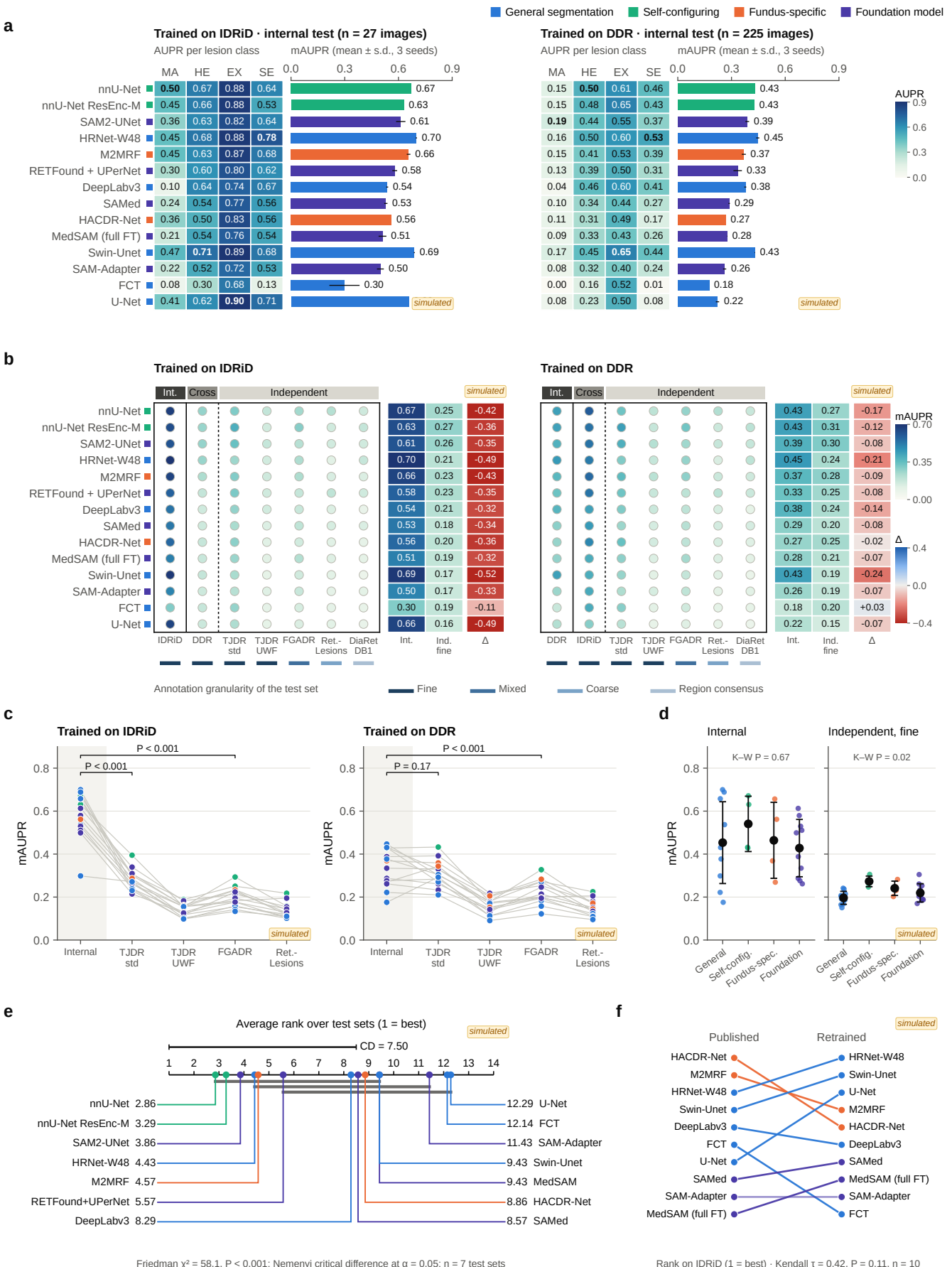

General segmentation
Self-configuring
Fundus-specific
Foundation model
a
Trained on IDRiD · internal test (n = 27 images)
AUPR per lesion class
mAUPR (mean ± s.d., 3 seeds)
MA HE EX SE
nnU-Net 0.50 0.67 0.88 0.64 0.67
nnU-Net ResEnc-M 0.45 0.66 0.88 0.53 0.63
SAM2-UNet 0.36 0.63 0.82 0.64 0.61
HRNet-W48 0.45 0.68 0.88 0.78 0.70
M2MRF 0.45 0.63 0.87 0.68 0.66
RETFound + UPerNet 0.30 0.60 0.80 0.62 0.58
DeepLabv3 0.10 0.64 0.74 0.67 0.54
SAMed 0.24 0.54 0.77 0.56 0.53
HACDR-Net 0.36 0.50 0.83 0.56 0.56
MedSAM (full FT) 0.21 0.54 0.76 0.54 0.51
Swin-Unet 0.47 0.71 0.89 0.68 0.69
SAM-Adapter 0.22 0.52 0.72 0.53 0.50
FCT 0.08 0.30 0.68 0.13 0.30
U-Net 0.41 0.62 0.90 0.71 0.69
simulated
Trained on DDR · internal test (n = 225 images)
AUPR per lesion class
mAUPR (mean ± s.d., 3 seeds)
MA HE EX SE
0.15 0.50 0.61 0.46 0.43
0.15 0.48 0.65 0.43 0.43
0.19 0.44 0.55 0.37 0.39
0.16 0.50 0.60 0.53 0.45
0.15 0.41 0.53 0.39 0.37
0.13 0.39 0.50 0.31 0.33
0.04 0.46 0.60 0.41 0.38
0.10 0.34 0.44 0.27 0.29
0.11 0.31 0.49 0.17 0.27
0.09 0.33 0.43 0.26 0.28
0.17 0.45 0.65 0.44 0.43
0.08 0.32 0.40 0.24 0.26
0.00 0.16 0.52 0.01 0.18
0.08 0.23 0.50 0.08 0.22
simulated
AUPR
b
Trained on IDRiD
Int. Cross Independent
IDRiD DDR TJDR std TJDR UWF FGADR Ret.-Lesions DiaRet DB1
Int. Ind. fine Δ
nnU-Net 0.67 0.25 -0.42
nnU-Net ResEnc-M 0.63 0.27 -0.36
SAM2-UNet 0.61 0.26 -0.35
HRNet-W48 0.70 0.21 -0.49
M2MRF 0.66 0.23 -0.43
RETFound + UPerNet 0.58 0.23 -0.35
DeepLabv3 0.54 0.21 -0.32
SAMed 0.53 0.18 -0.34
HACDR-Net 0.56 0.20 -0.36
MedSAM (full FT) 0.51 0.19 -0.32
Swin-Unet 0.69 0.17 -0.52
SAM-Adapter 0.50 0.17 -0.33
FCT 0.30 0.19 -0.11
U-Net 0.66 0.16 -0.49
simulated
Trained on DDR
Int. Cross Independent
DDR IDRiD TJDR std TJDR UWF FGADR Ret.-Lesions DiaRet DB1
Int. Ind. fine Δ
0.43 0.27 -0.17
0.43 0.31 -0.12
0.39 0.30 -0.08
0.45 0.24 -0.21
0.37 0.28 -0.09
0.33 0.25 -0.08
0.38 0.24 -0.14
0.29 0.20 -0.08
0.27 0.25 -0.02
0.28 0.21 -0.07
0.43 0.19 -0.24
0.26 0.19 -0.07
0.18 0.20 +0.03
0.22 0.15 -0.07
simulated
mAUPR
Δ
Annotation granularity of the test set
Fine
Mixed
Coarse
Region consensus
c
Trained on IDRiD
Trained on DDR
P < 0.001
P < 0.001
P < 0.001
P = 0.17
mAUPR
Internal
TJDR std
TJDR UWF
FGADR
Ret.-Lesions
simulated
d
Internal
Independent, fine
K–W P = 0.67
K–W P = 0.02
mAUPR
General
Self-config.
Fundus-spec.
Foundation
simulated
e
Average rank over test sets (1 = best)
CD = 7.50
simulated
nnU-Net 2.86
nnU-Net ResEnc-M 3.29
SAM2-UNet 3.86
HRNet-W48 4.43
M2MRF 4.57
RETFound+UPerNet 5.57
DeepLabv3 8.29
12.29 U-Net
12.14 FCT
11.43 SAM-Adapter
9.43 Swin-Unet
9.43 MedSAM
8.86 HACDR-Net
8.57 SAMed
f
Published
Retrained
HACDR-Net
M2MRF
HRNet-W48
Swin-Unet
DeepLabv3
FCT
U-Net
SAMed
SAM-Adapter
MedSAM (full FT)
HRNet-W48
Swin-Unet
U-Net
M2MRF
HACDR-Net
DeepLabv3
SAMed
MedSAM (full FT)
SAM-Adapter
FCT
simulated
Friedman χ² = 58.1, P < 0.001; Nemenyi critical difference at α = 0.05; n = 7 test sets
Rank on IDRiD (1 = best) · Kendall τ = 0.42, P = 0.11, n = 10